High Level Design (HLD)

Supply Chain Management

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Document Version Control

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10/03/2022	1.0	Abstract, Introduction, General Description	Shubham Tembhurne
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Abstract

AtliQ Mart is a growing FMCG (fast moving consumer Good) manufacturer headquartered in Gujarat, India. It is currently operational in three cities Vadodara, Surat and Ahmedabad. They want to expand to other metro/tier1 cities in the next 2 years.

1 Introduction

1. Why this High-Level Design Document?

The purpose of this High-Level Design (HLD) document is to add the necessary detail to the current project description to represent a suitable model for coding. This document is also intended to help detect contradictions before coding and can be used as a reference manual for how the modules interact at a high level.

The HLD will:

- Present all of the design aspects and define them in detail
- Describe the user interface being implemented
- Describe the hardware and software interfaces
- Describe the performance requirements
- Include design features and the architecture of the project
- List and describe the non-functional attributes like:

  -Security
  -Reliability
  -Maintainability
  -Portability
  -Reusability
  -Application compatibility
  -Resource utilization
  -Serviceability

1.2 Scope

The HLD documentation presents the structure of the system, such as the database architecture, application architecture (layers), application flow (Navigation), and technology architecture. The HLD uses non-technical to mildly-technical terms which should be understandable to the administrators of the system.

2 General Description

2.1 Product Perspective & Problem Statement

The goal of this project is to analyze to predict the probability of heart disease occurrence, based on a combination of features that describes the disease. To achieve the goal, we used a data set that is formed by taking into consideration some of the information of 303 individuals. The problem is based on the given information about each individual we have to calculate whether that individual will suffer from heart disease or not.

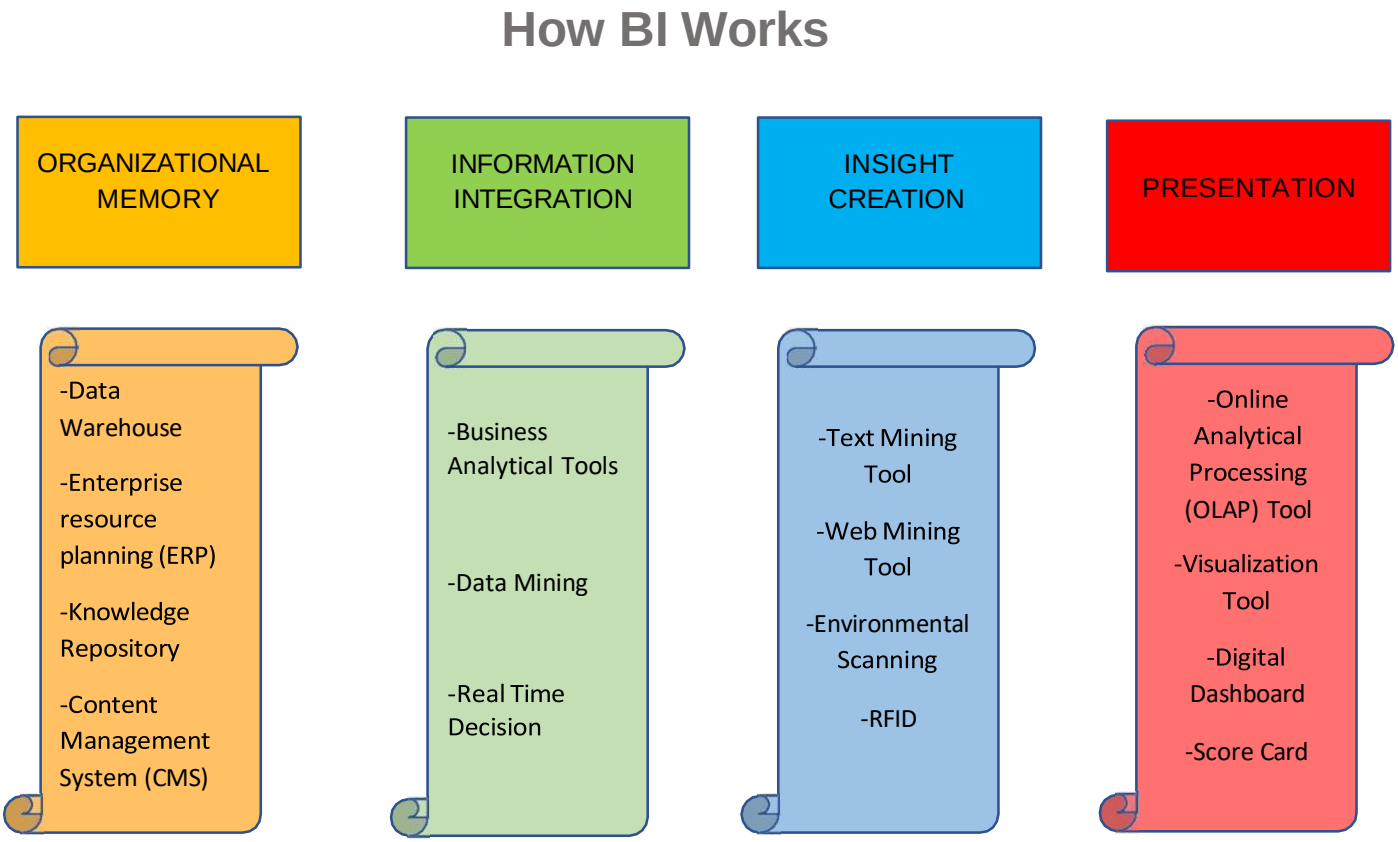
2.2 Tools used

Business Intelligence tools Power Bi, and Quadratic AI are used to build the whole framework.

3 Design Details

1. Functional Architecture

Figure 1: Functional Architecture of Business Intelligence



2. Optimization

1. Your data strategy drives performance

- Minimize the number of fields
- Minimize the number of records
- Optimize extracts to speed up future queries by materializing calculations, removing columns, and the use of accelerated views

2. Reduce the marks (data points) in your view

- Practice guided analytics. There's no need to fit everything you plan to show in a single view. Compile related views and connect them with action filters to travel from overview to highly-granular views at the speed of thought.

- Remove unneeded dimensions from the detail shelf.

- Explore. Try displaying your data in different types of views.

3. Limit your filters by number and type

- Reduce the number of filters in use. Excessive filters on a view will create a more complex query, which takes longer to return results. Double-check your filters and remove any that aren't necessary.

- Use an include filter. Exclude filters load the entire domain of a dimension while including filters do not. An include filter runs much faster than an exclude filter, especially for dimensions with many members.

- Use a continuous date filter. Continuous date filters (relative and range-of-date filters) can take advantage of the indexing properties in your database and are faster than discrete data filters.

- Use Boolean or numeric filters. Computers process integers and Booleans (t/f) much faster than strings.

- Use parameters and action filters. These reduce the query load (and work across data sources).

4. Optimize and materialize your calculations

- Perform calculations in the database
- Reduce the number of nested calculations.
- Reduce the granularity of LOD or table calculations in the view. The more granular the calculation, the longer it takes.

  ✓ LODs - Look at the number of unique dimension members in the calculation.

  ✓ Table Calculations - the more marks in the view, the longer it will take to calculate.

- Where possible, use MIN or MAX instead of AVG. AVG requires more processing than MIN or MAX. Often rows will be duplicated and display the same result with MIN, MAX, or AVG.

- Make groups with calculations. Like include filters, calculated groups load only named members of the domain, whereas Tableau's group function loads the entire domain.

- Use Booleans or numeric calculations instead of string calculations. Computers can process integers and Booleans (t/f) much faster than strings. Boolean>Int>Float>Date>DateTime>String.

4 KPI

Dashboards will be implemented to display and indicate certain KPIs and relevant indicators for the disease.

As and when the system starts to capture the historical/periodic data for a user, the dashboards will be included to display charts over time with progress on various indicators or factors

KPIs (Key Performance Indicators)

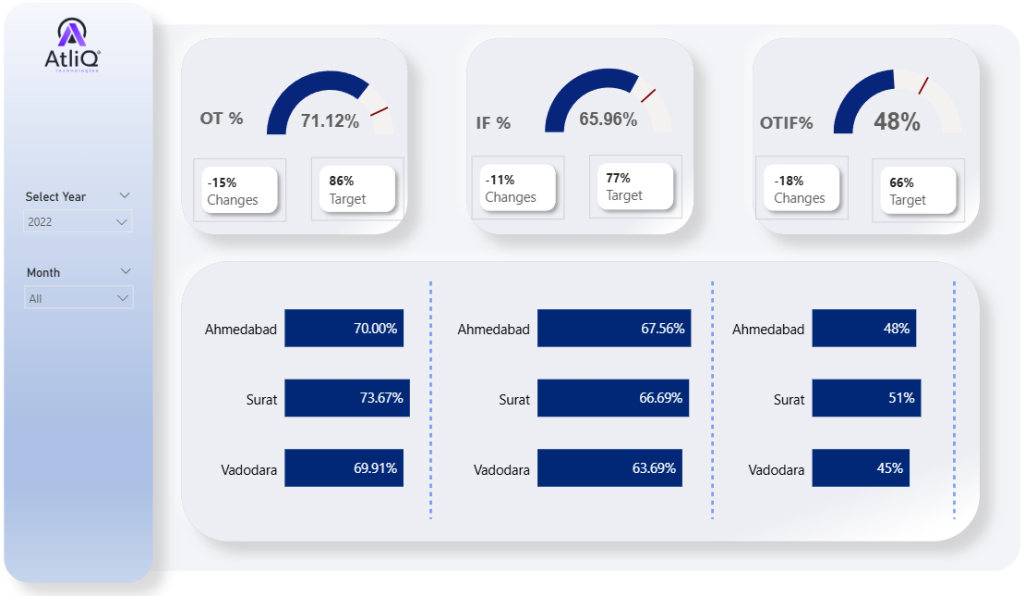
Key indicators displaying a summary of the Supply Chain and its relationship with different metrics

1. In Full%, On Time%, Otif% compares to target
2. Otif% by City vs target
3. Metrics by customers, products
4. Otif% by top Customers vs target
5. LIFR and VOFR for top 10 Product by order volume
6. LIFR and VOFR trends over time for top 5 products
7. LIFR and VOFR trends over time for Products category
8. Weekly order Quantity for top 5 Cities
9. Regional Performance summary
10. Delivery delay analysis

5 Deployment

Prioritizing data and analytics couldn't come at a better time. Your company, no matter what size, is already collecting data and most likely analyzing just a portion of it to solve business problems, gain competitive advantages, and drive enterprise transformation. With the explosive growth of enterprise data, database technologies, and the high demand for analytical skills, today's most effective IT organizations have shifted their focus to enabling self-service by deploying and operating Power BI at scale, as well as organizing, orchestrating, and unifying disparate sources of data for business users and experts alike to author and consume content.

Power BI prioritizes choice in flexibility to fit, rather than dictate, your enterprise architecture. Power BI Desktop and Power BI Service leverage your existing technology investments and integrate them into your IT infrastructure to provide a self-service, modern analytics platform for your users. With on-premises, cloud, and hosted options, there is a version of Power BI to match your requirements.



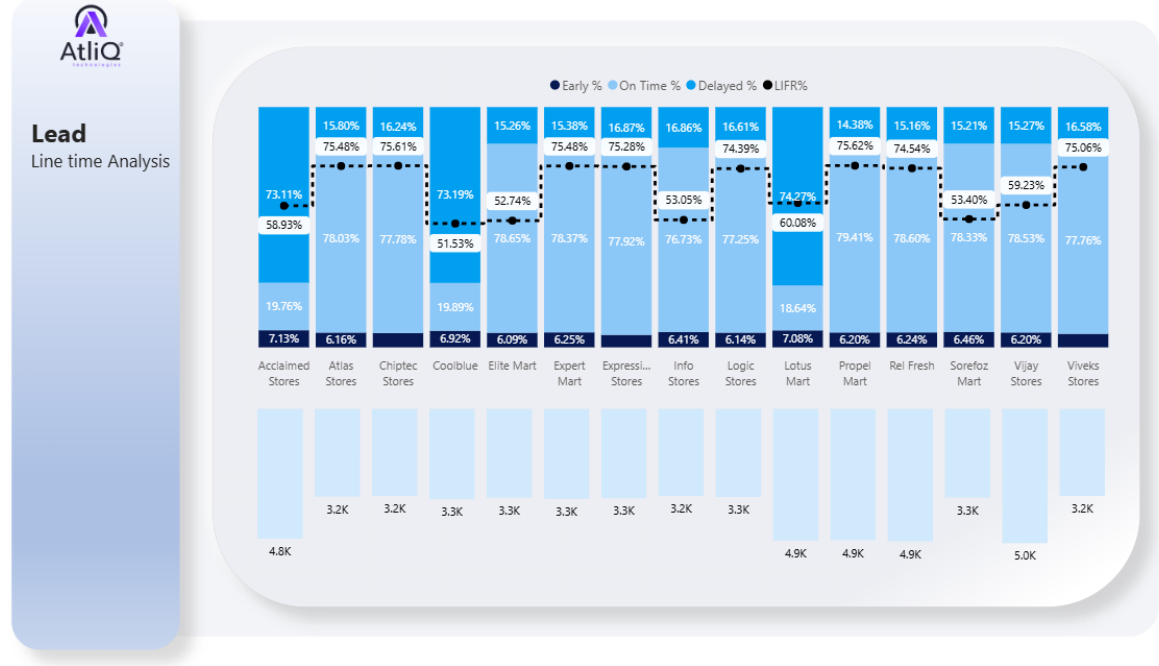
Metric	Acclaimed Stores	Atlas Stores	Chiptec Stores	Coolblue	Elite Mart	Expert Mart	Expression Stores	Info Stores	Logic Stores	Lotus Mart	Propel Mart	Rel Fresh	Sorefoz Mart	Vijay Stores	Viveks Stores
IF%	58.93%	75.48%	75.61%	51.53%	52.74%	75.48%	75.28%	53.05%	74.39%	60.08%	75.62%	74.54%	53.40%	59.23%	75.06%
OT%	26.89%	84.20%	83.76%	26.81%	84.74%	84.62%	83.13%	83.14%	83.39%	25.73%	85.62%	84.84%	84.79%	84.73%	83.42%
OTIF%	15.24%	63.80%	62.97%	13.75%	44.88%	63.61%	62.31%	43.48%	61.96%	16.06%	65.05%	63.04%	45.60%	49.61%	63.04%

AtliQ
Trends
for
Key Metrics
VOFR%
97.0%
96.5%
96.50%
96.54%
96.59%
96.66%
96.64%
96.74%
96.48%
96.37%
96.51%
96.70%
96.51%
96.58%
96.56%
96.75%
96.45%
96.59%
96.61%
96.39%
96.53%
96.53%
96.62%
96.88%
96.64%
96.67%
96.45%
96.74%
96.62%
W 10 W 11 W 12 W 13 W 14 W 15 W 16 W 17 W 18 W 19 W 20 W 21 W 22 W 23 W 24 W 25 W 26 W 27 W 28 W 29 W 30 W 31 W 32 W 33 W 34 W 35 W 36
96.85% 96.64% 96.87% 96.75% 96.80% 96.77%
96.56% 96.62% 96.56% 96.71% 96.70% 96.75%
96.32% 96.37% 96.35% 96.19% 96.37% 96.41%
March April May June July August
2022
Ahmedabad Surat Vadodara
96.68% 96.62% 96.62% 96.65% 96.65% 96.66%
96.58% 96.53% 96.61% 96.54% 96.57% 96.65%
96.55% 96.51% 96.59% 96.44% 96.56% 96.55%
March April May June July August
Beverages Dairy Food

AtliQ
Order Line
At Glance
57K
Total Order Line
12969K
Total Order LIne QTY
Deliverd
38K
Total Order Line In Full
13427K
Total Order LIne QTY
96.59%
VOFR%
65.96%
LIFR%
100%
50%
0%
75.06% 59.23% 53.40% 74.54% 75.62% 60.08% 74.39% 53.05% 75.28% 75.48% 52.74% 51.53% 75.61% 75.48% 58.93%
3.2K 5.0K 3.3K 4.9K 4.9K 4.9K 3.3K 3.2K 3.3K 3.3K 3.3K 3.3K 3.2K 3.2K 4.8K
Viveks Stores
Vijay Stores
Sorefoz Mart
Rel Fresh
Propel Mart
Lotus Mart
Logic Stores
Info Stores
Express... Stores
Expert Mart
Elite Mart
Coolblue
Chiptec Stores
Atlas Stores
Acclai... Stores
Total order Line LIFR%

Delayed Days	Beverages	Dairy	Food	Total
-1	584	2454	569	3607
0	6171	24521	6306	36998
1	1377	5557	1321	8255
2	843	3452	841	5136
3	486	2112	502	3100
Total	9461	38096	9539	57096

Delayed Days	Ahmedabad	Surat	Vadodara	Total
-1	1243	1094	1270	3607
0	12531	12050	12417	36998
1	2934	2389	2932	8255
2	1819	1458	1859	5136
3	1149	851	1100	3100
Total	19676	17842	19578	57096



category	product_name	LIFR%	VOFR%	LIFR% by Date	VOFR% by Date
Beverages	AM Tea 100	65.90%	96.63%		
	AM Tea 250	65.34%	96.52%		
	AM Tea 500	65.12%	96.48%		
Dairy	AM Butter 100	66.17%	96.62%		
	AM Butter 250	64.25%	96.38%		
	AM Butter 500	65.64%	96.59%		
	AM Curd 100	66.25%	96.58%		
	AM Curd 250	66.30%	96.61%		
	AM Curd 50	65.57%	96.58%		
	AM Ghee 100	65.96%	96.57%		
	AM Ghee 150	66.46%	96.68%		
	AM Ghee 250	65.50%	96.53%		
	AM Milk 100	66.46%	96.63%		
	AM Milk 250	65.63%	96.62%		
	AM Milk 500	66.85%	96.63%		
Food	AM Biscuits 250	66.41%	96.64%		
	AM Biscuits 500	66.74%	96.62%		
	AM Biscuits 750	66.73%	96.63%		